CS2600

THE UNIVERSITY OF WARWICK
Second Year Examinations, January 2016
Algorithms

- **Time allowed: 2 hours.**
- Answer **THREE** questions: **ONE** from Section A, and **TWO** from Section B.
- Make it clear on the cover of the answerbook which three questions you have selected.
- Calculators are not required and not permitted.

Continued

Section A

- Answer **ONE** question from this section.

CS2600

1. You are playing a 2D treasure-collecting game by operating a robot that starts in the bottom-left corner cell of an $m \times n$ array T (the cell of the array with coordinates $(1, 1)$), and that is supposed to finish in the top-right corner cell (the cell with coordinates (m, n)).

   At every move you may instruct the robot to move either up to the cell directly above, or to the cell directly to the right. Every cell with coordinates (i, j), where $1 \leq i \leq m$ and $1 \leq j \leq n$, has a treasure of value $T[i, j]$. The robot collects the treasures of all the cells that it visits.

   (a) **[5 marks]** Consider the following 4×3 array T of treasure values:

4	8	7	1
9	5	3	6
2	6	5	3

   For example, the bottom-left corner cell has a treasure of value $T[1, 1] = 2$ and the top-right corner cell has a treasure of value $T[4, 3] = 1$.

   What is the total value of all the treasures collected by the robot if it starts in the bottom-left corner cell and makes the following sequence of moves on its way to the top-right corner cell: right, right, up, right, up?

   (b) **[7 marks]** For the example array from part (a), what is the sequence of moves that you can instruct the robot to take from the bottom-left corner cell to the top-right corner cell, and that maximizes the total value of the treasures collected by the robot on the way.

   What is the maximum total of the treasures collected?

   (c) **[18 marks]** Design a polynomial-time algorithm that—given an $m \times n$ array T of treasure values—computes the maximum total value of treasures collected by the robot on the way from the bottom-left corner cell to the top-right corner cell, where only right and up moves are allowed.

   Illustrate your algorithm using the array T from part (a).

Continued

CS2600

2. You have n video files $F_1, F_2, \ldots, F_n$ that you want to store on your smartphone. For every $i = 1, 2, \ldots, n$, the file F_i has size $S[i]$ (say, it requires $S[i]$ megabytes of space). The capacity of the smartphone memory is C megabytes. Unfortunately, not all of the files can fit on the smartphone because $\sum_{i=1}^{n} S[i] > C$.

   (a) **[8 marks]** Suppose that you want to *maximize the number* of video files stored on the smartphone.

   Consider the greedy algorithm that selects the files in order of *increasing sizes*, and which stops just before exceeding the memory capacity C.

   Either prove that this algorithm is correct or provide a counter-example.

   (b) **[7 marks]** Suppose that you want to *use as much of the capacity* C of the smartphone memory as possible.

   Consider the greedy algorithm which selects files in order of *decreasing sizes*, and which stops just before exceeding the memory capacity C.

   Either prove that this algorithm is correct or provide a counter-example.

   (c) **[15 marks]** Assume that for every $i = 1, 2, \ldots, n$, the video file F_i has the star rating $R[i]$, which is an integer between 0 and 5, given by your favourite film critic. Suppose that you want to *maximize the sum of the star ratings* of the video files stored on the smartphone.

   Argue that the algorithmic problem of selecting such an optimal collection of files is a special case of the knapsack problem. State and justify the asymptotic running time of an algorithm for solving the knapsack problem as applied to the optimal file selection problem discussed here.

Continued

Section B

- Answer **TWO** questions from this section.

3. (a) **[4 marks]** A literal is a boolean variable or a negation of a boolean variable. A disjunctive normal form (DNF) formula is a disjunction of conjunctive clauses, where a conjunctive clause is a conjunction of literals.

   Give an example of a DNF formula that uses 4 boolean variables, and is a disjunction of 4 conjunctive clauses, each of which is a conjunction of 3 literals.

   Is the DNF formula that you have given satisfiable? If yes, give a satisfying assignment; if not, argue why not.

   (b) **[7 marks]** Provide a polynomial-time algorithm that, given a DNF formula, determines if it is satisfiable, and outputs a satisfying assignment if the answer is yes.

   (c) **[8 marks]** A path in a graph is simple if it does not visit any vertex more than once.

   The LONG-PATH decision problem is—given a directed graph G and given a number k—to determine if there is a simple path of length k in the graph G.

   Prove that LONG-PATH belongs to the complexity class NP.

   (d) **[8 marks]** The HAMILTONIAN-PATH decision problem is—given a directed graph G and given two vertices s and t in G—to determine if there is a simple path from s to t that visits every vertex in the graph G.

   Prove that the LONG-PATH problem is NP-complete; you may assume that HAMILTONIAN-PATH is NP-complete.

   (e) **[8 marks]** The 4DEG-INDEPENDENT-SET decision problem is—given an undirected graph G in which every vertex has degree at most 4, and given a number k—to determine if there is an independent set of size k in the graph G.

   The 2LIT-3SAT decision problem is—given a CNF boolean formula with at most three literals per clause, in which each literal occurs at most twice in the formula—to determine if the formula is satisfiable.

   Prove that 4DEG-INDEPENDENT-SET is NP-complete; you may assume that 2LIT-3SAT is NP-complete.

---

4. (a) **[4 marks]** Recall that $G = (V, E)$ is a bipartite graph if there is a partition of the set of vertices into disjoint sets L and R, such that every edge in E has one end in L and the other in R.

   Draw the bipartite graph in which $L = \{1, 2, 3, 4\}$, $R = \{5, 6, 7, 8\}$, and $E = \{(1, 5), (2, 5), (3, 5), (3, 6), (3, 8), (4, 6), (4, 7)\}$.

   (b) **[7 marks]** Recall that a *matching* in a graph $G = (V, E)$ is a set of edges in which no two distinct edges share a vertex.

   Give a matching of size 3 in the graph from part (a).

   Is there a matching of size 4 in the graph? If yes, give one; if not, give a brief but formal reason why not.

   (c) **[8 marks]** Describe a polynomial-time reduction from the problem of finding the maximum size of a matching in a bipartite graph to the problem of finding a maximum-value flow in a flow network.

   Illustrate the reduction on the bipartite graph from part (a) by drawing the corresponding flow network.

   (d) **[8 marks]** Consider the flow f in your flow network from part (c) that corresponds to the matching $\{(3, 5), (4, 6)\}$.

   Illustrate an iteration of the Ford-Fulkerson algorithm that starts from the flow f. Draw the residual graph G_f, indicate an augmenting path in it, and give the flow obtained from f by pushing an extra unit of flow along this path.

   Argue that the flow obtained by the above augmentation is a maximum-value flow by demonstrating an s-t cut whose capacity is equal to the value of the flow.

   (e) **[8 marks]** Is the following statement true or false?

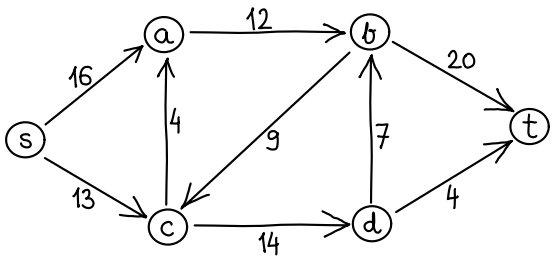
   > Every minimum s-t-cut in a flow network N is also a minimum s-t-cut in the network obtained from N by adding 1 to every edge capacity.

   If it is true, give a proof. If it is false, give a counter-example.

Continued

CS2600

5. (a) **[4 marks]** Consider the following flow network. The numbers labelling the edges are their capacities, and s and t are the source and sink vertices.



Find a flow of value 17 in the flow network.

State the flow conservation constraint for vertex b, and the capacity constraint for edge (d, b), and argue that they are satisfied for the flow that you have found.

(b) **[8 marks]** Find a flow f of maximum value in the flow network from part (a). Draw the residual graph G_f of the flow f that you have found, indicating the residual capacities of its edges.
Note: you do not need to show intermediate flows or their residual graphs.

(c) **[7 marks]** Prove that the flow that you have given in part (b) is indeed a flow of maximum value by showing an s-t cut in the network whose capacity is equal to the value of the flow.

(d) **[8 marks]** Let G be a flow network with integer edge capacities, and let f be a flow of maximum value in G. Provide an efficient algorithm that—given G, f, and an edge e in G—computes a flow of maximum value in the flow network obtained from G by decreasing the capacity of edge e by 1.

In order to obtain significant credit you should give an $O(|V| + |E|)$-time algorithm for the task.

(e) **[8 marks]** You are organizing a party and you want it to be as much fun as possible. You have n friends to choose from, but the maximum number of people that you can invite into your flat—its capacity—is C. The fun at a party stems from pairwise interactions between guests, and you have the $n \times n$ "fun table" F, such that for every pair (i, j) of your friends, $F[i, j]$ is a number between 0 and 1 that specifies how much fun their mutual presence at the party would contribute.

The DECENT-PARTY decision problem is—given an $n \times n$ array F representing the fun table, a number C and a number T as inputs—to determine if there is a selection of C guests such that the total fun contributed by all pairwise interactions between them is at least T.

Prove that if there is a polynomial-time algorithm for solving DECENT-PARTY then there is also one for the INDEPENDENT-SET decision problem.

End